

The rule of the numbers to be removed in the Sieve of Eratosthenes

Takamasa Ishikawa

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Abstract

The Sieve of Eratosthenes has a rule regarding the numbers that are removed. There is a cycle to the appearance of the numbers targeted for deletion and It becomes symmetrical. When the prime number under investigation is extremely large, the number required to reach the loop becomes enormous, yet it still remains within finite limits.

1 Introduction

I reasoned that if one could reliably calculate the number of primes smaller than a given number, it might be possible to determine whether an odd number is prime by examining the even numbers immediately preceding and following it. With the following method, the solution does not change even when crossing a prime number; therefore, no determination can be made.

- Divide by the two the value obtained by adding the one to or subtracting the one from the number.
- Divide that value by the three, and subtract the result from the original value.
- Perform the same calculation using the five.

We will proceed by considering each prime number in turn, continuing until the power of the prime exceeds the target number.

This calculation method does not yield an answer, as it is evident that the set includes identical values or that the intervals between the values are not uniform.

Based on the above, I began investigating the intervals between the numerical values targeted for deletion.

2 Procedures

The following describes the procedure for creating an array, removing elements that are multiples, and then reconstructing the array from the remaining elements. Therefore, we will refer to the respective arrangements as "list," "remaining," and "loop." Note: The "loop" is the same sequence as the "remaining" formed by preceding prime numbers.

Place the three at the beginning of the sequence

If you add the one to the odd numbers being examined and divide them by the two, all the resulting numbers will be odd. Insert a line break at multiples of the three and delete those numbers.

<u>line break</u>	<u>remove</u>
3, 5, 7	5, 7
9,11,13	11,13
15,17,19	17,19
21,23,25	23,25
27,29,31	29,31
33,35,37 →	35,37
39,41,43	41,43
45,47,49	47,49
51,53,55	53,55
57,59,61	59,61

...

Place the five at the beginning

If you first create a sequence starting with the five and insert a line break at every multiple of the five, the sequence begins to repeat the seven and the three. Looking at this list, you will notice a repeating pattern of the two $\times$ (the five - the one) in the "remaining" numbers—those remaining after multiples of the three have been removed.

<u>Line break</u>	<u>comparing the loop elements of prime the three</u>
5,7,11,13,17,19,23, 25,29,31,	→ 5,7,11,13,17,19,23,25,29,31, (list)
35,37,41,43,47,49,53, 55,59,61,	→ 35,37,41,43,47,49,53,55,59,61,
65,67, ...	→ 65,67, ...

A sequence of numbers with multiples of the five removed (loop)

7,11,13,17,19,23,29,31,

Next is the prime number the seven

You can automatically generate the list by adding the thirty to the "loop" numbers. This the thirty is obtained by multiplying the prime number the five by the difference between the prime numbers the three and the nine (i.e., $9 - 3 = 6$). Please repeat this process until it becomes the same number as the prime number.

7,11,13,17,19,23,29,31,
) + 30
 37,41,43,47,49,53,59,61,
) + 30
 67,71,73,77,79,83,89,91,
) + 30
 97,101,103,107,109,113,119,121,
) + 30
 127,131,133,137,139,143,149,151,
) + 30
 157,161,163,167,169,173,179,181,
) + 30
 187,191,193,197,199,203,209,211,

The intervals between multiples of the seven are 12, 7, 4, 7, 4, 7, 12, and 3. Their sum is the fifty six, which corresponds to the prime number the five (representing a count of the eight) multiplied by the seven. And the total number of "the seven"s removed is the eight. This is the same as the "remaining" number. A pattern can be observed in the difference between the leading values of each list: the two hundred ten is obtained by subtracting the seven from two hundred seventeen, while the thirty is obtained by subtracting the five from thirty five. This relationship corresponds to multiplying the current prime number by a difference associated with the immediately preceding prime number. You can create the list by adding this "difference multiplied by a prime number" to the next "loop".

Prime		Difference
5	2×3	6
7	6×5	30
11	30×7	210
13	210×11	2310
...	...	...

Line break

7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47,
 49, 53, 59, 61, 67, 71, 73,
 77, 79, 83, 89,
 91, 97,101,103,107,109,113
 119,121,127,131
 133,137,139,143,149,151,157
 161,163,167,169,173,179,181,187,191,193,197,199
 203,209,211

217,221,223,227,229,233,239,241,247,251,253,257,
 259,263,269,271,277,281 ..

↓

Loop

11,13,17,19,23,29,31,37,41,43,47,53,59,61,67,71,73,79,83,89,
97,101,103,107,109,113,121,127,131,137,139,143,149,151,157,
163,167,169,173,179,181,187,191,193,197,199,209,211,

221,223,227,229,233,239,241,247,251,253,257,263,269,271,277,281 ...

And the prime number the eleven

The interval until the loop for the prime number the eleven begins is the five hundred twenty eight, and the sum of the numbers removed is forty eight. This process unfolds as follows.

26,5, 10, 5, 11, 15, 4, 16, 10, 5, 9, 15, 16, 5, 15, 10, 5, 15, 11, 14, 20, 11, 5,
9,

5, 11, 20, 14, 11, 15, 5, 10, 15, 5, 16, 15, 9, 5, 10, 16, 4, 15, 11, 5, 10, 5, 26, 3

I would like to make a comparison using the sequence of intervals "12, 7, 4, 7, 4, 7, 12, 3," which corresponds to the prime number the seven. You can see the number "the three" at the end. And for the prime number the seven, the fourth "the seven" serves as the center, while for the prime number the eleven, the twenty fourth "the nine" acts as the center, creating symmetry.

The numbers to be removed are clearly multiples of that prime number, and when divided by that prime, they form the same sequence as the "remaining" sequence.

prime the seven remove number

multiples 49,77,91,119,133,161,203,217

divide 7,11,13,17,19,23,29,31

prime the eleven remove number

multiples 121,143,187,209,253,319,341,407,451,473,517,583,649,671,

divide 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47, 53, 59, 61,

multiples 737,781,803,869,913,979,1067,1111,1133,1177,1199,1243,1331,

divide 67, 71, 73, 79, 83, 89, 97, 101, 103, 107, 109, 113, 121,

multiples 1397,1441,1507,1529,1573,1639,1661,1727,1793,1837,1859,

divide 127, 131, 137, 139, 143, 149, 151, 157, 163, 167, 169,

multiples 1903,1969,1991,2057,2101,2123,2167,2189,2299,2321

divide 173, 179, 181, 187, 191, 193, 197, 199, 209, 211,

Care must be taken when removing multiples of prime numbers, because the list is constructed by adding differences to the multiples of prime numbers. The final value of the loop is the difference plus one.

List of corresponds to target number

If the target number is reached along the way, stop creating the list. Then, remove the numbers obtained by multiplying each value on the list by the initial value. The list of prime numbers is complete once the result of raising the number to a power exceeds the target value.

Summary

prime	remains	inflate	diffrence	lastNumber	center
5	2	8	6	31	15
7	8	48	30	211	105
11	48	480	210	2311	1155
13	480	5760	2310	30031	15015
17	5760	92160	30030	510511	255255
19	92160	1658880	510510	9699691	4849845
23	1658880	36495360	9699690	223092871	111546435
29	36495360	1021870080	223092870	6469693231	3234846615

Start list {2, 3, 5, 7}

type	Formura
Diference	$previousdifferecne \times currentprime$
Inflate	$previousremains \times (currentprime - 1)$
Total	$previousremains \times currentprime$
Remains	previous Inflate
Last number	$currentdifference \times currentprime + 1$

3 Proof

Interval the three immediately before the start of the loop

The fact that "the three" invariably appears just before the loop begins is due to the "remainders" that persist after removing multiples from the "list" of the prime number the five. The "interval until the next number is removed" is the two because twenty five was removed from the "list" of primes (starting with the five) generated by the loop for the prime the three. Therefore, since the "list" are positioned according to the current count of prime numbers, the remainder of "the two" is placed before the start of the next "list". The first number in the "list" is a multiple of the prime number in question, and the two numbers falling within the range of twice that prime remain without being excluded.

Symmetrical

The reason for this symmetry, too, lies in the loop associated with the prime number "the five". Collapse eight of the items in the list, leaving the remaining two.

prime the five				
✓				
13,	11,	7,	5	
17,	19	,23	,25	,29 ,31,
↓				
4,	8,	16,	20,	

Then, you notice that the multiples of the five to be removed are located in symmetrical positions. The boundary line where that value falls is "the fifteen." This figure represents half of the difference—"6 × 5 = 30"—calculated at the start of the loop.

prime the seven

			✓				✓						
103	101	97	91	89	83	79	77	73	71	67	61	59	53
107	109	113	119	121	127	131	133	137	139	143	149	151	157
								↓					
4	8	16	28	32	44	52	46	64	68	76	88	92	104
✓													✓
49	47	43	41	37	31	29	23	19	17	13	11	7	
161	163	167	169	173	179	181	187	191	193	197	199	203	
			↓										
112	116	124	128	136	148	152	164	172	176	184	188	196	
remainder 209,211													

prime the eleven

1153	1151	1147	1139	1129	1123	1121	1117	...
1157	1159	1163	1171	1181	1187	1189	1193	...
					↓			
4	8	16	32	52	64	68	76	...
...	1093	1091	1087		.907	901	899	...
...	1217	1219	1223		1403	1409	1411	...
		↓					↓	
	124	128	136		496	508	512	

The difference between the central number and the number symmetrical to it is constant, and the sum of each pair is equal to the difference between the starting numbers of the loops. Although not directly related to the numbers being removed, subtracting the vertically aligned numbers reveals a binary sequence. While not yet fully verified, it appears that as the prime number being examined increases, the binary sequence appears more continuously. Based on the explanation above, the remaining columns are arranged in a pattern of "mirror image, remaining 2." Since the list is constructed by repeating this pattern according to the prime number, the mirror image is invariably positioned at the center.

rimaining	list
symmetry, 2	
...	
symmetry, 2	
symmetry, 2	→ symmetry, 2, symmetry, 2, symmetry, 2, symmetry, 2, symmetry,
symmetry, 2	remainder 2,
...	
symmetry, 2	

In conclusion, since the original array is symmetric, the resulting array is also symmetric.

A proof that holds true infinitely

Based on the explanation regarding the persistence of the three and the generation of symmetry, the shape of the sequence is maintained as long as the same procedure is repeated. If it can be demonstrated that values already excluded—or their multiples—will not appear in the future, that serves as evidence. I explained earlier that the list is created by adding the difference to the remaining values. The difference is the product of all prime numbers smaller than the current prime number. For example, the difference for the eleven is the two hundred ten, which is expressed as $2 \times 3 \times 5 \times 7$. It is divisible by prime numbers less than the eleven, but the number obtained by adding the one to it is not. When excluding all prime numbers smaller than the current prime number, as well as their multiples. The "remaining" number is not divisible by any number less than or equal to the current prime number. Therefore, neither numbers less than the current prime number nor numbers divisible as multiples of that prime number will appear.

$$q = \{p_1 \times p_2 \times p_3 \times \dots p(n-1)\} + p_n's \text{ "remaining"}$$
$$q/p_1 \text{ or } p_2 \text{ or } p_3 \dots p(n-1) = a + remainder$$

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The programs and related datasets used to verify the results of this study are publicly available at:

GitHub: https://github.com/TakamasaIshikawa/Rule_of_remove_num_in_Sieve_of_Eratosthenes

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